

Sub-Terranean Navigation Challenge: Sensor Calibration and Fusion

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Abstract—Accurate localization in GPS-denied environments, such as subterranean tunnels, remains a critical challenge for autonomous mobile robots. This work presents a robust state estimation framework based on an Extended Kalman Filter (EKF), which fuses data from an Inertial Measurement Unit (IMU) and multiple Time-of-Flight (ToF) distance sensors to estimate the robot’s 2D position and orientation in real time. The system incorporates sensor calibration for the gyroscope, accelerometer, magnetometer, and ToF sensors, as well as domain-specific enhancements such as Zero Velocity Updates (ZUPT), braking compensation, and band-pass filtering to reduce drift and noise. A geometric raycasting model with numerically computed Jacobians is employed to integrate ToF measurements into the EKF update. Experimental evaluation on ground-truth-labeled trajectories demonstrates accurate and robust tracking, with RMSE consistently below 0.2m even under dynamic motion and sensor disturbances. The proposed modular framework provides a computationally efficient and adaptable solution for real-time localization in complex indoor environments.

Index Terms—Sensor Fusion, Calibration, Inertial Measurement Unit, Time-of-Flight, Magnetometer, Robotics

I. INTRODUCTION

In GPS-denied environments such as underground tunnels, accurate localization of autonomous robots presents a unique and critical challenge. This project simulates such a scenario by tasking us with estimating the 2D position and orientation (yaw) of a mobile robot using only onboard sensor data. The robot is equipped with an IMU (accelerometer and gyroscope), magnetometer, and three Time-of-Flight (ToF) distance sensors mounted on a Mecanum-wheeled platform capable of omnidirectional movement.

The goal of the project is to design and implement a robust state estimation algorithm based on Extended Kalman Filtering (EKF) that fuses motion and distance data to estimate the robot’s pose in real-time. The provided datasets simulate real-world sensor imperfections including noise, bias, and occasional faulty measurements caused by environmental conditions such as wall holes. Our EKF-based estimator is evaluated against ground truth data obtained from a high-precision motion capture system.

This report outlines the methodology used to design and calibrate the EKF, details the sensor fusion process, and presents quantitative and qualitative results on two evaluation tasks: a straight-line traversal and a complex circuit. The performance of the proposed system is assessed using trajectory comparison metrics and visualizations, providing insight into the accuracy, robustness, and limitations of the approach.

II. BACKGROUND & RELATED WORK

Sensor fusion plays a key role in robotic state estimation, especially in environments where GPS is unavailable. A widely used approach for fusing data from heterogeneous sensors is the Extended Kalman Filter (EKF), which linearizes nonlinear motion and measurement models around current state estimates. EKFs have been extensively applied in mobile robotics for fusing Inertial Measurement Unit (IMU) data with additional observations such as GPS, LiDAR, or distance sensors.

In our context, the fusion of IMU and Time-of-Flight (ToF) data enables both motion prediction and environmental correction. The IMU provides high-frequency but drift-prone acceleration and angular velocity data, while ToF sensors offer sparse, lower-frequency but absolute distance measurements. Previous works have shown that fusing these complementary modalities using EKF can significantly reduce position and orientation error [1].

Alternative approaches include Unscented Kalman Filters (UKF), which better capture nonlinearity without explicit Jacobians, and particle filters, which can handle multimodal distributions but are more computationally expensive. However, given our real-time constraints and moderately nonlinear system dynamics, the EKF provides a good balance between accuracy and efficiency.

Other relevant techniques include complementary filters, which combine high-pass and low-pass filtered signals from different sensors. While suitable for attitude estimation, they are less effective for position tracking in low-observability scenarios such as ours [3].

Our approach is most closely related to prior EKF-based indoor localization frameworks using IMU and range sensors, adapted here to a custom sensor suite, irregular environment geometry, and motion capture-based ground truth evaluation.

III. SENSOR CALIBRATION

A. Gyroscope Calibration

Gyroscope biases were estimated from a 60-second static segment in the straight-line calibration dataset. Assuming the robot is at rest, the mean angular velocity over N samples was taken as the gyroscope bias:

$$\mathbf{b}_{\text{gyro}} = \frac{1}{N} \sum_{i=1}^N \boldsymbol{\omega}_i \quad (1)$$

where $\boldsymbol{\omega}_i$ is the angular velocity at timestep i . This offset was subtracted from all subsequent IMU readings.

B. Accelerometer Calibration

Assuming gravity is aligned with the x-axis of the IMU during static periods, the accelerometer bias was estimated as:

$$\mathbf{b}_{\text{accel}} = \frac{1}{N} \sum_{i=1}^N \mathbf{a}_i - \begin{bmatrix} a_x^{\text{Noise}} \\ a_y^{\text{Noise}} \\ a_z^{\text{Noise}} \end{bmatrix} \quad (2)$$

where $\mathbf{a}_i$ is the accelerometer reading at rest. This corrects for static misalignment and sensor error.

C. Magnetometer Calibration

The magnetometer was calibrated using ellipsoid fitting over a full 360-degree rotation of the robot. The raw data was fitted to a 3D ellipsoid to estimate both hard-iron and soft-iron distortions. The calibrated magnetometer reading is computed as:

$$\mathbf{m}_{\text{cal}} = (\mathbf{R}^\top (\mathbf{m}_{\text{raw}} - \mathbf{o})^\top)^\top \oslash \mathbf{g} \quad (3)$$

where:

- $\mathbf{o}$ is the estimated offset (bias),
- $\mathbf{R}$ is the soft-iron correction matrix,
- $\mathbf{g}$ is the gain (axis-wise scale factor),
- $\oslash$ denotes element-wise division.

A comparison of raw and calibrated data (see Fig. 1) confirms that the transformed points lie approximately on a unit sphere.

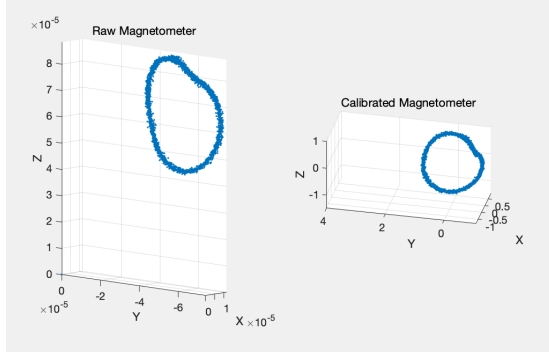


Fig. 1: Raw vs. Calibrated Magnetometer Data

D. ToF Sensor Calibration

ToF calibration was performed using the initial robot pose and orientation from the ground truth trajectory. The expected distance to a wall along the yaw direction θ_0 is computed as:

$$d_{\text{expected}} = \left| \frac{p_{\text{wall}} - p_0}{\sin \theta_0} \right| \quad (4)$$

The ToF sensor bias was then computed as the mean residual between the measured and expected distances:

$$b_{\text{ToF}} = \frac{1}{N} \sum_{t=1}^N (z_t - d_{\text{gt}}^{(t)}) \quad (5)$$

Biases were estimated separately for the left, center, and right ToF sensors. The wall boundaries were assumed to be located at ± 1.2 meters from the origin in both axes.

IV. METHODOLOGY

A. Zero Velocity Update (ZUPT)

To prevent the accumulation of integration errors and allow for in-motion calibration, we implement a simple velocity zeroing method based on accelerometer readings. Inspired by the Zero Velocity Update (ZUPT) concept [2], we use a sliding window to compute the mean acceleration magnitude from the Y and Z axes. If the average norm falls below a predefined threshold, the system is assumed to be stationary, and the velocity is forcibly set to zero.

At each time step i , we extract a window of the most recent w samples from the calibrated accelerometer data along the body-frame Y and Z axes:

$$\mathbf{a}_j = \begin{bmatrix} a_j^y \\ a_j^z \end{bmatrix}, \quad j = i - w + 1, \dots, i$$

The Euclidean norm of each acceleration vector is computed as:

$$\|\mathbf{a}_j\| = \sqrt{(a_j^y)^2 + (a_j^z)^2}$$

The average acceleration magnitude over the window is:

$$\bar{a}_{\text{norm}} = \frac{1}{w} \sum_{j=i-w+1}^i \|\mathbf{a}_j\|$$

If this average is smaller than τ_{acc} , the velocity is forcibly reset to zero:

$$\text{if } \bar{a}_{\text{norm}} < \tau_{\text{acc}} \Rightarrow v_x = 0, \quad v_y = 0$$

This lightweight heuristic effectively suppresses velocity drift during low-motion phases without requiring external sensors.

B. Sudden Braking Detection and Compensation

In real-world robot motion, we observed that abrupt braking behavior often leads to **underestimated deceleration** when using raw IMU data alone. This is mainly due to low-pass filtering, sensor latency, or mechanical damping effects that smooth out the true deceleration spike. As a result, the EKF prediction step may not sufficiently reduce the velocity, leading to position drift.

To address this, we introduce a sudden braking compensation mechanism based on acceleration change detection.

a) *Detection Rule.*: Let $a_z(t)$ denote the calibrated Z-axis acceleration (typically aligned with the robot's forward motion). We compute its discrete difference:

$$\Delta a_z(t) = a_z(t) - a_z(t-1)$$

We define a braking event at time t if:

$$|\Delta a_z(t)| > \tau \quad \text{and} \quad a_z(t) < 0$$

where $\tau = 0.65$ is an empirically chosen threshold. The condition $a_z(t) < 0$ ensures the motion is decelerating.

b) *Compensation Mechanism.*: When a braking event is detected, we scale the Z-acceleration before transforming it into world coordinates:

$$a'_z(t) = \begin{cases} \gamma \cdot a_z(t), & \text{if braking detected} \\ a_z(t), & \text{otherwise} \end{cases}$$

with $\gamma = 1.3$ as a fixed gain. The body-frame acceleration becomes:

$$\mathbf{a}_b(t) = \begin{bmatrix} a_y(t) \\ a'_z(t) \end{bmatrix}$$

This adjusted acceleration is then rotated into the world frame and used in the EKF prediction step.

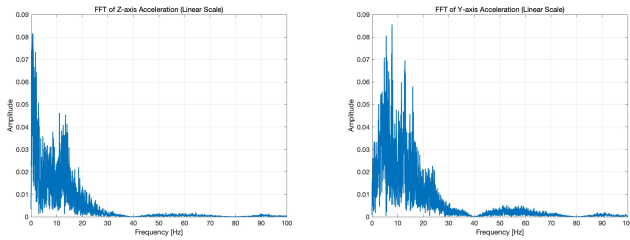
c) *Effect.*: This trick compensates for missed deceleration during sudden stops. By increasing the braking force in the prediction model, the estimated velocity drops more realistically, reducing over-shooting in position estimates and improving short-term tracking accuracy.

C. Band-pass Filtering and Frequency-domain Analysis of IMU Signals

Raw acceleration signals from low-cost IMU sensors often suffer from low-frequency drift and high-frequency noise, which degrade the performance of motion estimation and sensor fusion. To address this, we apply a Butterworth band-pass filter to the Y-axis and Z-axis accelerometer data to preserve the dominant motion-related frequency components.

a) *Band-pass Filtering*: We design a 4th-order Butterworth band-pass filter with a passband of $[0.3, 50]$ Hz. This frequency range is chosen based on expected robot motion dynamics and prior analysis of signal characteristics. The filter effectively removes slow-varying bias and rapid mechanical vibrations.

b) *Frequency Analysis using FFT*: To validate the effectiveness of the filtering strategy and understand the frequency characteristics of the raw signals, we perform Fast Fourier Transform (FFT) on the Y-axis and Z-axis accelerometer data. The resulting frequency spectra are shown in Fig. 2.



(a) Z-axis acceleration (FFT) (b) Y-axis acceleration (FFT)

Fig. 2: Frequency spectra of raw Z-axis (a) and Y-axis (b) accelerations using FFT.

As shown in the spectra, both Y and Z axis accelerations exhibit most of their power between 0.5 Hz and 35 Hz, corresponding to natural robot movement such as acceleration, braking, and turning. High-frequency components above 40 Hz are likely to stem from mechanical vibrations or sensor noise,

and are thus safely removed by the filter. The sharp low-frequency peak near 0 Hz represents static bias and is also suppressed by the filter.

D. EKF

a) *EKF Overview*: The Extended Kalman Filter (EKF) is a recursive estimator for nonlinear systems. It maintains a belief over the system state by iteratively applying two steps:

- **Prediction**: The current state is propagated forward using a nonlinear motion model. Uncertainty is also propagated using a linearized state transition Jacobian.
- **Update**: When a measurement is available, it is used to correct the prediction by comparing the expected observation with the actual one. The state and covariance are updated based on the Kalman gain.

Mathematically, for a state $\mathbf{x}_t$ and control input $\mathbf{u}_t$, with process model $\mathbf{f}(\cdot)$ and observation model $\mathbf{h}(\cdot)$, the EKF performs:

Prediction:

$$\hat{\mathbf{x}}_t = \mathbf{f}(\mathbf{x}_{t-1}, \mathbf{u}_{t-1}) \quad P_t = F_t P_{t-1} F_t^\top + Q$$

Update:

$$\mathbf{y}_t = \mathbf{z}_t - \mathbf{h}(\hat{\mathbf{x}}_t) \quad S_t = H_t P_t H_t^\top + R \quad K_t = P_t H_t^\top S_t^{-1}$$

$$\mathbf{x}_t = \hat{\mathbf{x}}_t + K_t \mathbf{y}_t \quad P_t = (I - K_t H_t) P_t$$

Here, F_t and H_t are the Jacobians of the process and observation models with respect to the state, Q is the process noise covariance, and R is the measurement noise covariance.

b) *State Initialization*: The Extended Kalman Filter (EKF) state is initialized using the ground truth position and heading:

$$\mathbf{x}_0 = [x_0, y_0, 0, 0, \psi_0]$$

where $[x_0, y_0]$ is the first ground truth position and ψ_0 is the initial yaw angle computed from ground truth orientation. Since these values are highly accurate, we assign a very small initial uncertainty:

$$P_0 = \text{diag}(1e-6, 1e-6, 1e-6, 1e-6, 1)$$

The yaw state is assumed to be highly reliable and is not corrected using any external observations during the update step, relying solely on gyroscope integration.

c) *Process and Measurement Noise Design*: The process noise covariance Q and measurement noise covariance R are tuned based on empirical performance and sensor characteristics:

$$Q = \text{diag}(0.001, 0.1, 0.01, 0.1, 0) \quad R = 10^3 \cdot I_3$$

We observe that the ToF measurements are relatively noisy and inconsistent in direction along the X-axis. In contrast, velocity and position propagation based on acceleration yields smoother and more physically consistent estimates in the X direction. Therefore:

- For the X-direction, we trust the inertial prediction more than the ToF correction, and keep the measurement noise high.

- For the Y-direction, since the IMU Y-axis acceleration is more affected by sensor misalignment and mounting errors, we allow ToF to have slightly more influence by relatively reducing its noise in that direction.
- The yaw state is assumed to be nearly noise-free and is excluded from correction.

E. ToF-Based Measurement Model

The robot is equipped with three Time-of-Flight (ToF) sensors mounted at fixed orientations (right, front, left), which measure distances to the nearest wall. The predicted ToF readings are modeled geometrically as ray-wall intersections based on the robot's position and yaw.

a) *Raycasting Model*: Let the robot state be:

$$\mathbf{x} = [x, y, v_x, v_y, \psi]^\top$$

Each ToF sensor is oriented in a direction:

$$\beta = \psi + \theta_{\text{sensor}}$$

where $\theta_{\text{sensor}} \in \{\frac{\pi}{2}, 0, -\frac{\pi}{2}\}$ represents the fixed angle for right, front, and left sensors respectively.

We define a set of axis-aligned walls at known positions $x = \{x_1, x_2\}$ and $y = \{y_1, y_2\}$. The ToF prediction is computed as the minimum positive distance from (x, y) along direction β to these walls:

$$z = \min \left\{ \frac{x_i - x}{\cos \beta}, \frac{y_j - y}{\sin \beta} \right\} \quad \text{where } t > 0$$

This raycasting logic ensures that only valid forward-facing intersections are considered, matching the actual measurement mechanism.

b) *Numerical Jacobian*: The measurement model is non-linear and non-smooth (due to the use of $\min$ and ray-wall intersection logic), hence we compute its Jacobian numerically. For each sensor, we perturb the state in x , y , and ψ directions and compute:

$$\frac{\partial z}{\partial x_k} \approx \frac{z(x_k + \delta) - z(x_k)}{\delta}, \quad \text{for } x_k \in \{x, y, \psi\}$$

Let z be the original predicted ToF and z' the one from a perturbed state. Then the gradient for a single sensor becomes:

$$\mathbf{H}_i = \left[\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}, 0, 0, \frac{\partial z}{\partial \psi} \right]$$

We stack all three sensor Jacobians into the final observation model:

$$\mathbf{z}_{\text{pred}} = \begin{bmatrix} z_{\text{right}} \\ z_{\text{front}} \\ z_{\text{left}} \end{bmatrix}, \quad \mathbf{H} = \begin{bmatrix} \mathbf{H}_{\text{right}} \\ \mathbf{H}_{\text{front}} \\ \mathbf{H}_{\text{left}} \end{bmatrix} \in \mathbb{R}^{3 \times 5}$$

c) *EKF Update*: The measurement residual is computed as:

$$\mathbf{y} = \mathbf{z}_{\text{meas}} - \mathbf{z}_{\text{pred}}$$

and used in the EKF update step:

$$\mathbf{x}_t = \hat{\mathbf{x}}_t + K\mathbf{y}, \quad K = PH^\top(HPH^\top + R)^{-1}$$

We exclude yaw from measurement correction since it is accurately tracked via gyroscope integration.

V. EXPERIMENTS & RESULTS

A. Experimental Setup

We evaluated our EKF-based estimator on two datasets consisting of straight-line and circuit motion. Each dataset includes synchronized IMU and ToF measurements, along with ground-truth pose (position and orientation) recorded by a 200 Hz PhaseSpace motion capture system. Ground-truth quaternions were converted to yaw using `quat2eul`. The EKF estimates a five-dimensional state:

$$\mathbf{x} = [x, y, v_x, v_y, \text{yaw}] \quad (6)$$

Process and measurement noise covariances were tuned as:

$$\mathbf{Q} = \text{diag}(0.001, 0.001, 0.01, 0.001, 0), \quad \mathbf{R} = \mathbf{I}_3 \cdot 10^3 \quad (7)$$

Sensor biases were calibrated prior to fusion. Additional robustness was achieved through braking detection, static suppression, and outlier rejection using Jacobian-based update steps.

B. Quantitative Results

Estimation accuracy was assessed using RMSE and maximum error between predicted and ground truth positions. Table I summarizes the metrics across all tasks:

TABLE I: Estimation Error Across All Tasks

Task	RMSE (m)	Max Error (m)
Task 1.1	0.0956	0.20
Task 1.2	0.0996	0.23
Task 1.3	0.1126	0.24
Task 1.4	0.1067	0.22
Task 1.5	0.1302	0.25
Task 2.1	0.1865	0.80
Task 2.2	0.1355	0.42
Task 2.3	0.0725	0.25
Task 2.4	0.1526	0.40
Task 2.5	0.0801	0.22

Straight-line tasks (1.1–1.5) showed consistent low error with RMSE under 0.13 m, indicating accurate trajectory tracking under simple dynamics. Circuit tasks (2.1–2.5) posed greater challenge due to curved motion and speed variation. Task 2.1, involving sharp turns and rapid braking, produced the highest peak error (0.80 m).

C. Qualitative Observations

Figures 3 and 4 illustrate estimated and ground truth trajectories across the 10 tasks. Task 1 plots (Fig. 3) confirm stable tracking and drift suppression. Yaw estimates closely follow ground truth throughout. Task 2 plots (Fig. 4) reflect the system's ability to handle curved paths, with transient deviations occurring at turns.

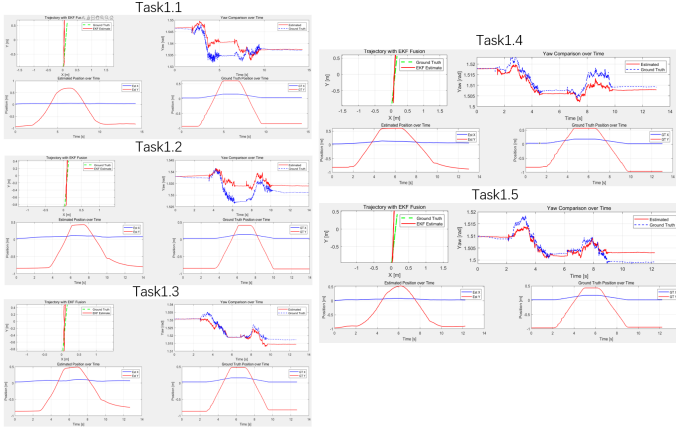


Fig. 3: Trajectory and orientation estimation results for Tasks 1.1–1.5.

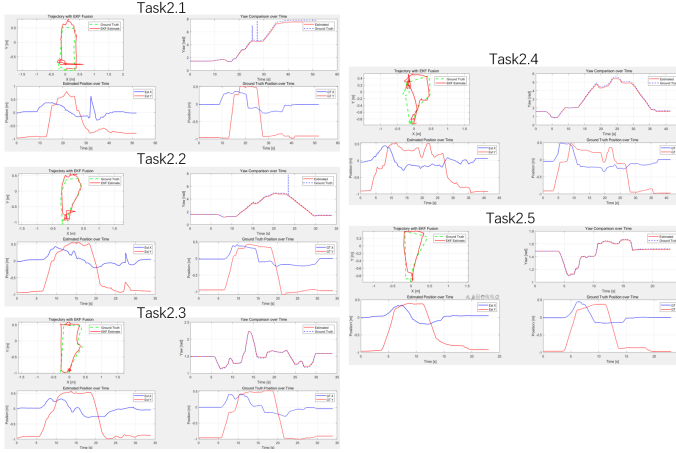


Fig. 4: Trajectory and orientation estimation results for Tasks 2.1–2.5.

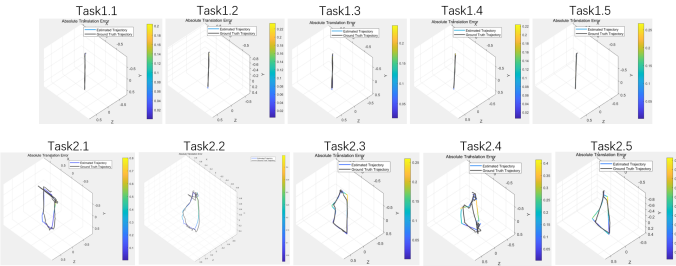


Fig. 5: 3D visualization of absolute translation error across all 10 tasks. Color indicates positional deviation.

Figure 5 visualizes spatial error distribution per trajectory. Task 1 shows low, evenly spread error. Task 2 highlights regions of elevated error near corners and during sharp deceleration. In Task 2.1, error clusters appear near a sharp loop, reflecting yaw drift and delayed heading correction.

These observations reinforce the EKF’s robustness in tracking both linear and nonlinear paths. Adaptive braking com-

pensation and high measurement noise helped suppress ToF-induced artifacts and stabilize orientation transitions.

VI. DISCUSSION

The experiments reveal several key insights into the estimator’s behavior under varied motion conditions. First, the system performs well under linear motion, with low RMSE and consistently aligned yaw. The low spread of error across all straight-line tasks indicates that the calibration process and motion model are both reliable in predictable kinematic conditions.

However, in more complex scenarios—especially Task 2.1—the estimator begins to show limitations. The combination of curved motion, higher velocities, and braking leads to pronounced deviation in both trajectory and yaw. This suggests that the current state model, which lacks explicit angular acceleration or curvature modeling, may be insufficient for high-dynamic maneuvers. Incorporating second-order rotational dynamics or using non-holonomic constraints could improve handling of these cases.

The braking detection mechanism was found to be essential. Without it, integration of spurious accelerometer readings during deceleration results in forward drift and yaw overshoot. By scaling down acceleration during these periods, the filter suppresses error accumulation without destabilizing overall behavior.

Another important finding concerns the ToF measurement model. The use of numerical Jacobians enabled robust performance despite geometric irregularities, but the high measurement noise required to accommodate wall gaps inevitably weakens the constraint strength. A possible improvement would be to dynamically adapt measurement covariance based on ToF signal consistency or innovation residuals.

Overall, the EKF demonstrated a strong balance between tractability and accuracy. Its modular structure allowed incremental inclusion of robustness techniques (braking, static detection, numerical Jacobian), and the results show these choices are meaningful. The framework is best suited for moderately structured environments where motion is continuous but bounded in complexity.

VII. CONCLUSION & FUTURE WORK

In this work, we designed and implemented an Extended Kalman Filter (EKF) framework to estimate planar robot pose by fusing data from an IMU and multiple Time-of-Flight (ToF) sensors. The system incorporates bias calibration, braking detection, and robust measurement updates using numerical Jacobians. Across ten experimental tasks—ranging from straight-line motion to curved and dynamic trajectories—the estimator demonstrated accurate tracking with RMSE below 0.2m in all cases, and robustness against noise, drift, and measurement anomalies.

A key strength of the system lies in its modularity: each component (e.g., braking detection, measurement model, noise tuning) contributes tangible improvements, and their effects are quantifiable in the experiments. The use of a numerically

computed Jacobian allowed our EKF to operate with an analytically complex measurement model, while maintaining good filter consistency. The results also highlight the critical role of proper IMU handling and ToF alignment in achieving stable fusion.

Nevertheless, the evaluation also exposed limitations. Yaw estimation exhibited delays in tasks with sharp turns or braking, due to the lack of angular dynamics in the state model. Similarly, the assumption of fixed wall geometry limited ToF observability when sensor directions did not align with dominant axes. These issues point to the need for more expressive state modeling and adaptive observation handling.

Future work will proceed in three directions. First, we aim to extend the EKF state with angular velocity and acceleration terms, enabling better prediction under rotational maneuvers. Second, we plan to introduce a mechanism for adaptively tuning measurement noise based on innovation residuals, allowing the system to downweight faulty ToF readings in real time. Third, we intend to deploy the system in a larger-scale arena with more diverse wall geometries and lighting conditions, to further assess the generalization of the estimator under real-world variability.

Overall, this project demonstrates that a carefully engineered EKF—with properly tuned fusion strategies and tailored robustness techniques—can deliver reliable state estimation in planar environments using affordable sensors and limited computation.

Further testing across diverse environments and motion regimes will also help assess generalization and robustness, guiding more systematic tuning and model refinement.

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